

Rafael D. De Paz
Independent Researcher
<https://logos.pub>

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Editorial Board
Foundations of Physics
Springer

Dear Editors,

I am writing to submit the enclosed manuscript, entitled

**“Information-Theoretic Gravity:
Proton Mass from Vacuum Geometry and the
Emergence of the Schwarzschild Metric,”**

for consideration for publication in *Foundations of Physics*.

Summary. The paper presents a derivation of gravity as an emergent informational phenomenon. Using the holographic mass framework of Hramein, Guernonprez, and Alirol (2023), we derive the proton rest mass from the quantum vacuum energy density through a two-surface screening mechanism containing *zero free parameters*. The predicted proton charge radius ($r_p = 0.84132$ fm) agrees with CODATA 2018 to better than 0.01%. We prove that the Schwarzschild metric emerges naturally from the first screening surface and provide an independent topological derivation via Gauss-Codazzi embedding in S^3 .

Suggested Reviewers.

1. **Prof. Erik Verlinde** (University of Amsterdam) — Author of the entropic gravity framework.
2. **Prof. Gerard 't Hooft** (Utrecht University) — Pioneer of the holographic principle.
3. **Prof. Leonard Susskind** (Stanford University) — Co-developer of the holographic principle and string landscape.

This manuscript has not been previously published and is not under consideration at any

other journal. I confirm no conflicts of interest.

Respectfully submitted,

Rafael D. De Paz
Independent Researcher
hi@logos.pub